

Tide retrospective: the leading-part instantiation

Tide loop (Claude Fable 5, with GPT-5.6 Sol deliberation)

2026-08-09

1 Setting

Ninth tide of the germbij arc, and the last item on its follow-up list: the multivariate analogue of the analytic instantiation.¹ The multivariate sector bound consumes a scaled set of directions with an amplitude bound at small scales; in the note this comes from a nonzero leading homogeneous part of the difference of losses. Mathlib’s analytic order theory is one-variable only, so the multivariate factorisation is taken as the natural hypothesis pair (a homogeneous leading part and a remainder of one higher order, which is what a multivariate Taylor expansion supplies), and the tide turns that pair into the sector data.

2 The thing formalised

One theorem, `leading_part_scaled_set` in `Laplace/Multi/LeadingPart.lean`, sorry-free with zero warnings. If P is continuous, homogeneous of degree m under nonnegative scalings, nonzero at some x_0 with $\|x_0\| = 3/2$, and $|a - P| \leq C\|x\|^{m+1}$ near 0, then there are a closed ball S around x_0 inside the sup-norm annulus (measurable, of positive finite volume), $c > 0$ and $u_0 > 0$ with

$$cu^m \leq |a(ux)| \quad \text{for all } x \in S, u \in (0, u_0].$$

This is exactly the data `sector_lower_bound_multi` consumes, so the multivariate chain now runs from leading-part structure just as the 1D chain runs from analyticity.

3 Proof strategy

Continuity keeps $|P|$ above half its center value on a small ball (the same `eventually_gt_nhds` pattern as the 1D instantiation); the ball radius is capped at $1/2$ so S sits inside $\|x\| \leq 2$; homogeneity turns $|P(ux)|$ into $u^m|P(x)|$; and the scale threshold u_0 is chosen so the remainder $C(2u)^{m+1}$ eats at most half the leading term. The volume facts are `measure_ball_pos` and `measure_closedBall_lt_top`.

4 Roadblocks and resolutions

Three compile iterations, all lint-level or naming: a rewrite against a `set`-abbreviation that the goal no longer mentioned (replaced by the direct term), the triangle-inequality step (the sum-form lemma has moved; the difference form `abs_sub_abs_le_abs_sub` plus `abs_sub_comm` is stable), and

¹2026-08-09-02-55-tide-germbij-analytic.

one long line. Operationally, a parallel session was discovered mid-tide running its own tides on this repo (a new branch appeared on fetch, and its rebase output leaked into this session's shell through the shared `.git`); the worktree discipline kept the two sessions from interfering, and `main`'s reflog was verified clean.

5 What was Mathlib, what was new

Mathlib: continuity/filter idioms, ball measures, power monotonicity. New: the statement design, in particular taking the Taylor pair (homogeneous part, remainder bound) as the interface where one-variable proofs could lean on `analyticOrderAt`: the right hypothesis shape for multivariate callers until Mathlib grows multivariate order theory.

6 Lessons

When Mathlib lacks a factorisation theory (multivariate analytic order), take the factorisation's *output* as the hypothesis rather than blocking on upstream gaps; the statement stays honest and the caller (a Taylor expansion) can discharge it. And in shared-`.git` setups, expect other sessions' git chatter in your shell output; verify with reflog rather than assuming damage.

7 Follow-ups

- The multivariate turnkey composite: leading-part instantiation + `sector_lower_bound_multi` + the multivariate composite, mirroring the 1D corollary/turnkey pair.
- A bridging lemma from multivariate Taylor expansions (e.g. `ContDiff` at order $m+1$) to the remainder hypothesis, when a concrete caller wants it.
- Note annotation: bump the pin and tag Lemma 7.2's scaled-set discussion once merged.